Internet Protocol (IP) and TCP/IP Model — Detailed Theory

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1. Internet Protocol (IP)

Overview

- IP is the principal communications protocol in the Internet protocol suite for relaying datagrams (packets) across network boundaries.
- Its main function is to deliver packets from the source host to the destination host based on their IP addresses.
- IP defines the addressing system and packet structure and operates at the **Network Layer (Layer 3)** of the OSI model.

IP Versions

- **IPv4:** The most widely used version, uses 32-bit addresses (e.g., 192.168.1.1), supporting approximately 4.3 billion unique addresses.

- **IPv6:** The newer version developed to overcome IPv4 address exhaustion. It uses 128-bit addresses (e.g., 2001:0db8::1), supporting a vastly larger address space.

IP Addressing

- IP addresses uniquely identify devices on a network.
- An IPv4 address consists of 4 octets (8 bits each).
- Address classes (A, B, C, D, E) existed but now subnetting and CIDR are used for flexible allocation.
- IPv6 addresses are hexadecimal, divided into eight 16-bit blocks.

IP Packet Structure (IPv4)

- **Header:** Contains fields such as Version, Header Length, Type of Service (TOS), Total Length, Identification, Flags, Fragment Offset, Time to Live (TTL), Protocol, Header Checksum, Source IP, Destination IP, and optional fields.

- **Payload:** The actual data being carried, which may be a TCP segment, UDP datagram, or other protocols.

IP Functions

- **Routing:** IP packets are forwarded by routers based on destination IP addresses.
- **Fragmentation and Reassembly:** Large packets are broken down into smaller fragments to traverse networks with smaller Maximum Transmission Units (MTUs).
- **Addressing:** Provides a logical address to identify devices across networks.

Key Concepts

- **TTL (Time to Live):** Limits the lifespan of a packet to prevent it from circulating indefinitely.
- **Protocol Field:** Identifies the protocol (e.g., TCP=6, UDP=17) carried in the payload.

- **Checksum:** Error-checking of the IP header to ensure integrity.

IPv6 Improvements

- Larger address space (128-bit)
- Simplified header for faster processing
- No fragmentation by routers (done by sender)
- Built-in support for IPsec (security)
- Improved multicast and anycast addressing

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2. TCP/IP Model

Overview

- The **TCP/IP model** (also known as the Internet protocol suite) is a conceptual framework for computer network protocols.
- Developed in the 1970s, it underpins the internet and most modern networks.
- It is simpler than the OSI model, with 4 layers instead of 7.

TCP/IP Model Layers and Their Functions

Layer Name	Function & Protocol Examples	OSI Layer Equivalent
Application	Provides network services to applications. Protocols: HTTP, FTP, SMTP, DNS, SSH	OSI Layers 5-7 (Session, Presentation, Application)
Transport	Provides reliable/unreliable data transfer. Protocols: TCP, UDP	OSI Layer 4 (Transport)
Internet	Handles logical addressing and routing of packets. Protocol: IP (IPv4/IPv6), ICMP, ARP	OSI Layer 3 (Network)

Network Access	Manages physical transmission of data and network hardware. Protocols: Ethernet, Wi-Fi, ARP	OSI Layers 1-2 (Physical, Data Link)

Key Protocols in TCP/IP Model

- **IP (Internet Protocol):** Provides addressing and routing.
- **TCP (Transmission Control Protocol):** Connection-oriented, reliable data transmission with error checking, flow control, and sequencing.
- **UDP (User Datagram Protocol):** Connectionless, minimal overhead, used for streaming, DNS.
- **ICMP (Internet Control Message Protocol):** For network diagnostics (e.g., ping).
- **ARP (Address Resolution Protocol):** Maps IP addresses to MAC addresses on local networks.

- **DNS (Domain Name System):** Resolves domain names to IP addresses.

TCP vs UDP

- **TCP:** Ensures data is received accurately and in order via acknowledgments, retransmissions, and congestion control.
- **UDP:** Sends datagrams without guarantees, used where speed is prioritized over reliability.

How TCP/IP Works in Communication

1. Application layer protocols generate data.
2. Transport layer (TCP/UDP) segments the data and adds headers.
3. Internet layer (IP) adds IP headers for routing.

4. Network Access layer encapsulates data into frames for physical transmission.

5. At the receiver, layers reverse the process to reconstruct data.

Summary of Differences between OSI and TCP/IP Models

Feature	OSI Model	TCP/IP Model
Number of layers	7	4
Layer naming	Physical, Data Link, Network, Transport, Session, Presentation, Application	Network Access, Internet, Transport, Application
Development basis	Theoretical, general-purpose	Practical, Internet-focused
Layer combination	Separate Session and Presentation	Combined into Application layer